Q1: [5+5]Show that if n is an integer and $n^3 + 5$ is odd, then n is even using
**a)** a proof by contraposition.
**b)** a proof by contradiction.

**(a) Proof by Contraposition**

The contrapositive of "If n^3 + 5 is odd, then n is even" is:
If n is **odd**, then n^3 + 5 is **even**.
Assume n is odd. Then:
n = 2k + 1 for some integer k
Compute n^3:
n^3 = (2k+1)^3 = 8k^3 + 12k^2 + 6k + 1
So:
n^3 = 2(4k^3 + 6k^2 + 3k) + 1
Hence, n^3 is **odd**.
n^3 + 5 = odd + odd = even

**(b) Proof by Contradiction**

Negating the given statement: ~(p->q) ≡ ~(~(p ∨)q) ≡ p ∧ ~q

n^3 + 5 is odd, and n is odd (negation of "if $n^3 + 5$ is odd, then n is even")

n is even then n = 2k + 1 by definition:

n^3 + 5 = (2k+1)^3 + 5

(2k+1)^3 = 8k^3 + 12k^2 + 6k + 1

n^3 + 5 = 8k^3 + 12k^2 + 6k + 1 + 5 = 8k^3 + 12k^2 + 6k + 6

= 2(4k^3 + 6k^2 + 3k + 3)

We assumed n^3 + 5 is odd, but derived that it is even.
This is a contradiction.

Q2: [5] If an integer n > 4 is a perfect square, then the integer immediately preceding it (n−1) is not prime.

Since n is a perfect square, there exists an integer k \ge 3 such that
n = k^2
Now consider:
n - 1 = k^2 - 1
Factor using difference of squares:
k^2 - 1 = (k - 1)(k + 1)

- Since k > 3, we have: k - 1 > 2 k + 1 > 4
- Thus, n - 1 = (k - 1)(k + 1) is a product of two integers greater than 1.

Q3: [10] At a certain university 2/3 of the mathematics students and 3/5 of the computer science students have taken a discrete mathematics course. The number of mathematics students who have taken the course equals the number of computer science students who have taken the course. If there are at least 100 mathematics students at the university, what are the least possible number of mathematics students and the least possible number of computer science students at the university?

Let:

- M = number of mathematics students
- C = number of computer science students
- Mathematics students who took the course: 2/3M
- Computer science students who took the course: 3/5C
-

Given: 2/3M=3/5C

10M = 9C
M = 9/10C or C = 10/9M

Let:= M = 9k, C = 10k for some integer k.
Apply constraint M >= 100
9k >= 100 → k >= 100/9 ≈ 11.11
Smallest integer k = 12

- M = 9 x12 = 108
- C = 10 x 12 = 120
- 2/3 x 108 = 72
- 3/5 x 120 = 72